

Algebra 1
Unit 2
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

1. The **quotient of powers law** of exponents states keep the then _____ the powers.

add

subtract

multiply

base

coefficient

exponent

and

2. The **power of a quotient law** of exponents states the exponential expression $\left(\frac{a}{b}\right)^m$ is equal to

☐ a^{b-m} .

☒ $\frac{a^m}{b^m}$.

☐ $a^m - b^m$.

Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

3. $\frac{-4x^4y^2}{10xy^7}$

$$\frac{-2x^3}{5y^5}$$

4. $\frac{3a^{10}b^6}{6b^2c}$

$$\frac{a^{10}b^4}{2c}$$

5. $\left(\frac{-2r^3s}{3t^4}\right)^6$

$$\frac{64r^{18}s^6}{729t^{24}}$$

6. $\frac{-k^7j^3}{7k^9j^3}$

$$-\frac{1}{7k^2}$$

$$7. \quad \frac{46w^3z^{10}}{33x^2y^{11}}$$

$$\frac{46w^3z^{10}}{33x^2y^{11}}$$

$$8. \quad \frac{-144c^9d^2e^0}{98c^4d^{11}e}$$

$$\frac{-72c^5}{49d^9e}$$

$$9. \quad \left(\frac{z^0}{-3x^3}\right)^4$$

$$\frac{1}{81x^{12}}$$

$$10. \quad \frac{5s^{15}t^6}{-10s^{12}t}$$

$$-\frac{s^3t^5}{2}$$